# Cortical Dynamical System Equations (TPModel)

April 20, 2026

The cortical model used in this repository is a stochastic, discrete-time, linear state-space system:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k, \tag{1}$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{x}_k. \tag{2}$$

Here,

$$\mathbf{x}_k \in \mathbb{R}^n, \quad \mathbf{u}_k \in \mathbb{R}^m, \quad \mathbf{y}_k \in \mathbb{R}^{(10n)^2}. \tag{3}$$

## State Matrix Construction

$$\mathbf{A} = \mathbf{A}_{\text{base}} + \kappa\,\widetilde{\mathbf{W}}. \tag{4}$$

$\mathbf{A}_{\text{base}}$ is block-diagonal with damped rotation blocks

$$\mathbf{R}(\theta, \delta) = \delta \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{5}$$

Distance-based coupling is row-normalized:

$$\widetilde{W}_{ij} \propto \exp\left(-\frac{d_{ij}^2}{2\sigma_c^2}\right), \quad \widetilde{W}_{ii} = 0, \quad \sum_j \widetilde{W}_{ij} = 1. \tag{6}$$

## Process Noise

Spatial covariance:

$$Q_{ij} = \sigma_w^2 \exp\left(-\frac{d_{ij}^2}{2\sigma_Q^2}\right), \tag{7}$$

with a small diagonal regularization term added in code for positive definiteness.

Temporal coloring uses AR(1):

$$\mathbf{z}_k = \alpha\mathbf{z}_{k-1} + \sqrt{1-\alpha^2}\,\boldsymbol{\varepsilon}_k, \quad \boldsymbol{\varepsilon}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{8}$$

$$\mathbf{w}_k = \mathbf{L}\mathbf{z}_k, \quad \mathbf{L}\mathbf{L}^\top = \mathbf{Q}. \tag{9}$$

## Image Observation Model

Each state component modulates a Gaussian spatial blob in the cortical image:

$$C_{p,i} = \exp\left(-\frac{(x_p - x_i)^2 + (y_p - y_i)^2}{2\sigma_b^2}\right), \tag{10}$$

where pixel index $p$ corresponds to a location $(x_p, y_p)$, and $(x_i, y_i)$ is the center of cortical area $i$.

## Input Matrix Construction

For channel $j$ targeting area $t_j$, input spread is Gaussian in cortical distance:

$$B_{ij} \leftarrow 0.05 \, \exp\left(-\frac{d_{i,t_j}^2}{2\sigma_u^2}\right) \text{ (normalized)}, \tag{11}$$

$$B_{t_j,j} \leftarrow 0.30. \tag{12}$$